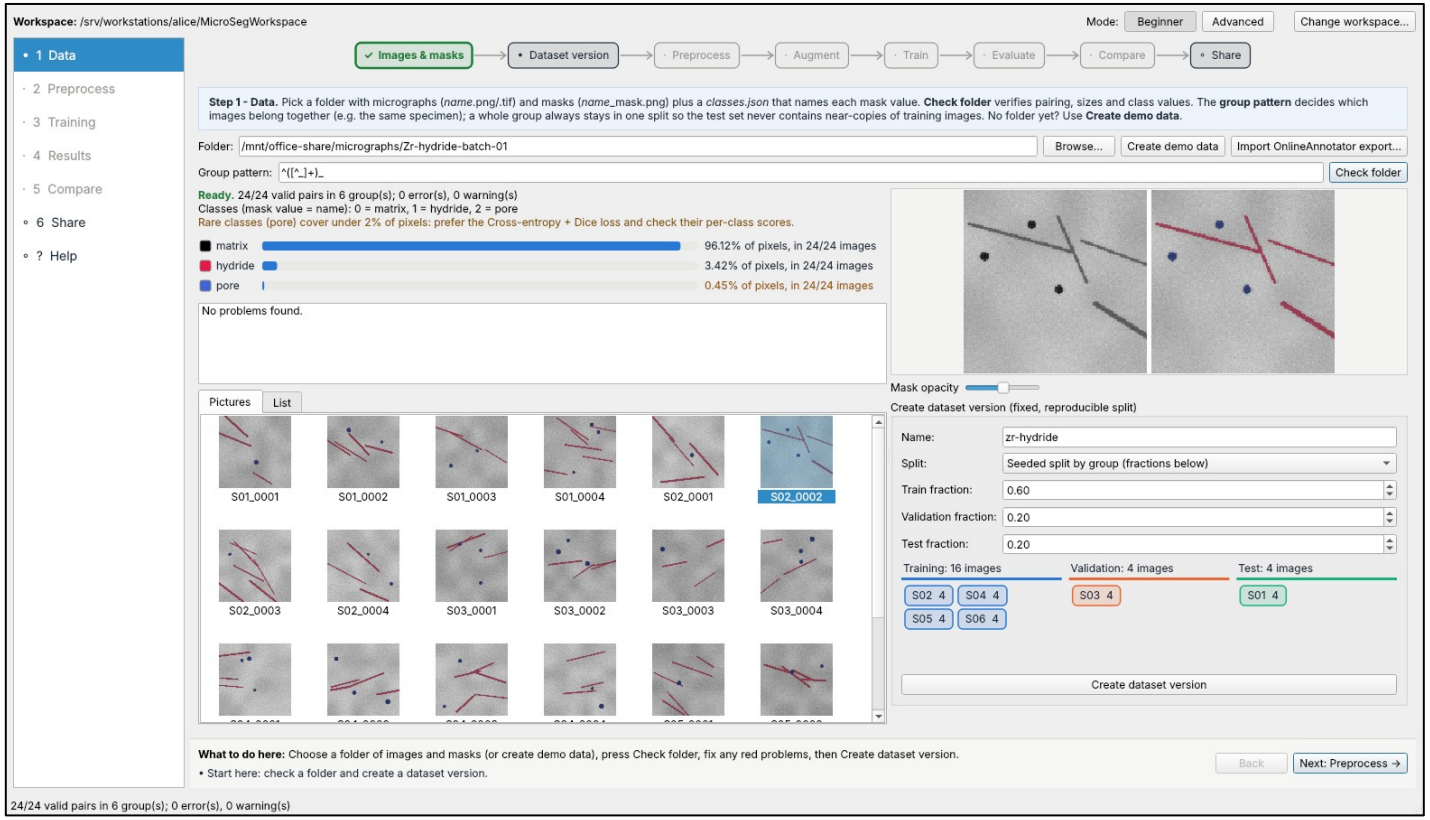
MicroSegWorkbench: a beginner's guide

Train, check, compare and share microstructure segmentation models

For researchers with no programming background
Version 1.0 release candidate · November 2026

About the screenshots
Real app, real training on CPU. Images are synthetic; a local folder (/mnt/office-share) stands in for the office network drive.



Simulated data and network drive

Six guided steps; the app checks your work at every step

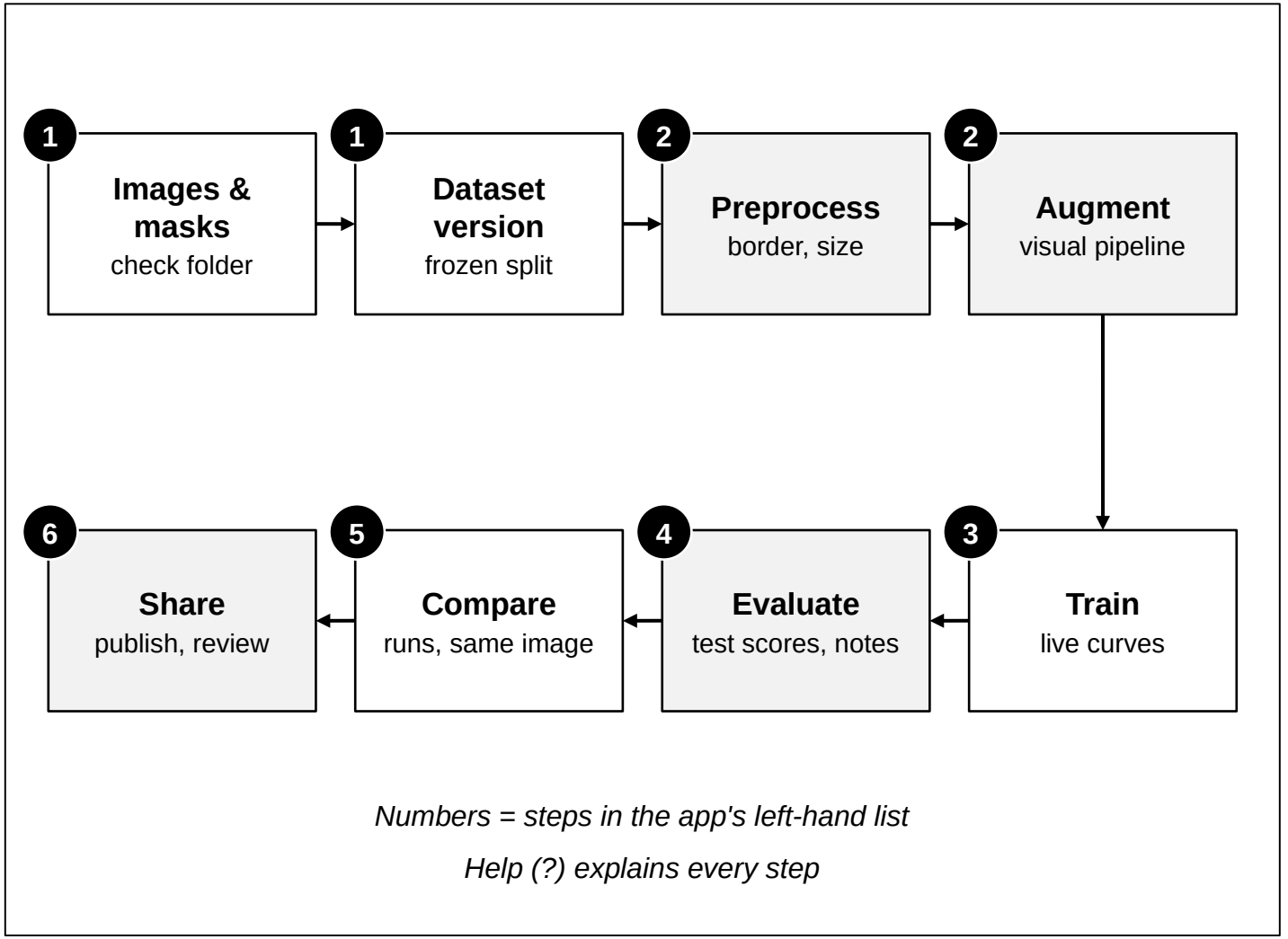
What the workbench does

Trains segmentation networks
on your annotated micrographs, on a GPU or the CPU

Keeps every result traceable
to exact images, settings, network and software

Checks your work at each step
data, settings, training and results, in plain words

Shares through the office drive
a colleague reviews before a model counts as approved



One guided path from annotated micrographs to an approved model

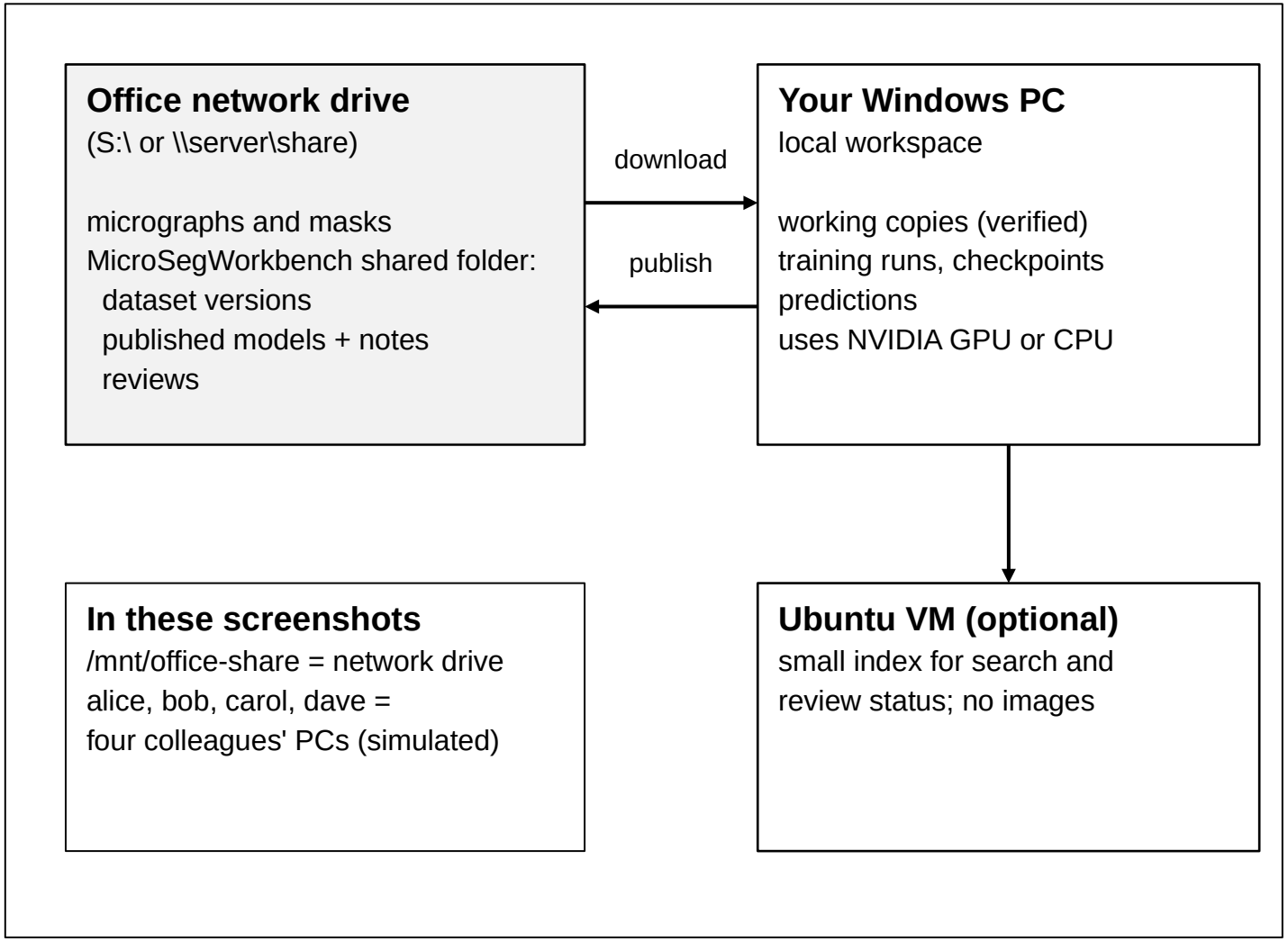
Where your files live

Network drive: the authoritative copy
micrographs, dataset versions, published models, reviews

Your PC: the workspace
verified local copies, runs in progress, caches

Publishing is checked
files are fingerprinted, never overwritten, never half-written

ATTENTION
Never move, rename or edit files inside the workspace or the shared folder by hand. Use the app.



Originals stay on the network drive; your PC works on verified copies

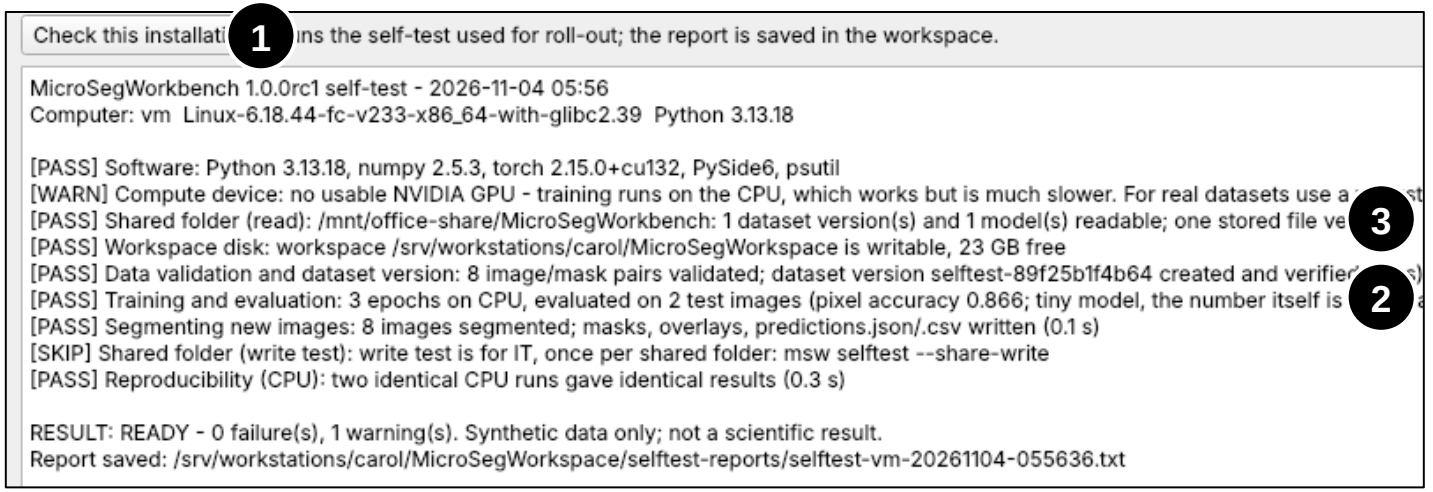
Check the installation first

1. **Help -> Check this installation**
   or msw selftest; about a minute

2. **Real training on tiny synthetic data**
   GPU, disk, data, training, evaluation, segmentation, reproducibility

3. **Shared folder: read check**
   nothing is written; IT tests publishing once per share (--share-write)

Report saved for sign-off
workspace\selftest-reports



CPU-only sandbox: the GPU check shows WARN

! **ATTENTION**
NOT READY? Each FAIL line names the step and the reason; send the report to IT.

A READY self-test on every workstation before real work

Finding your way around

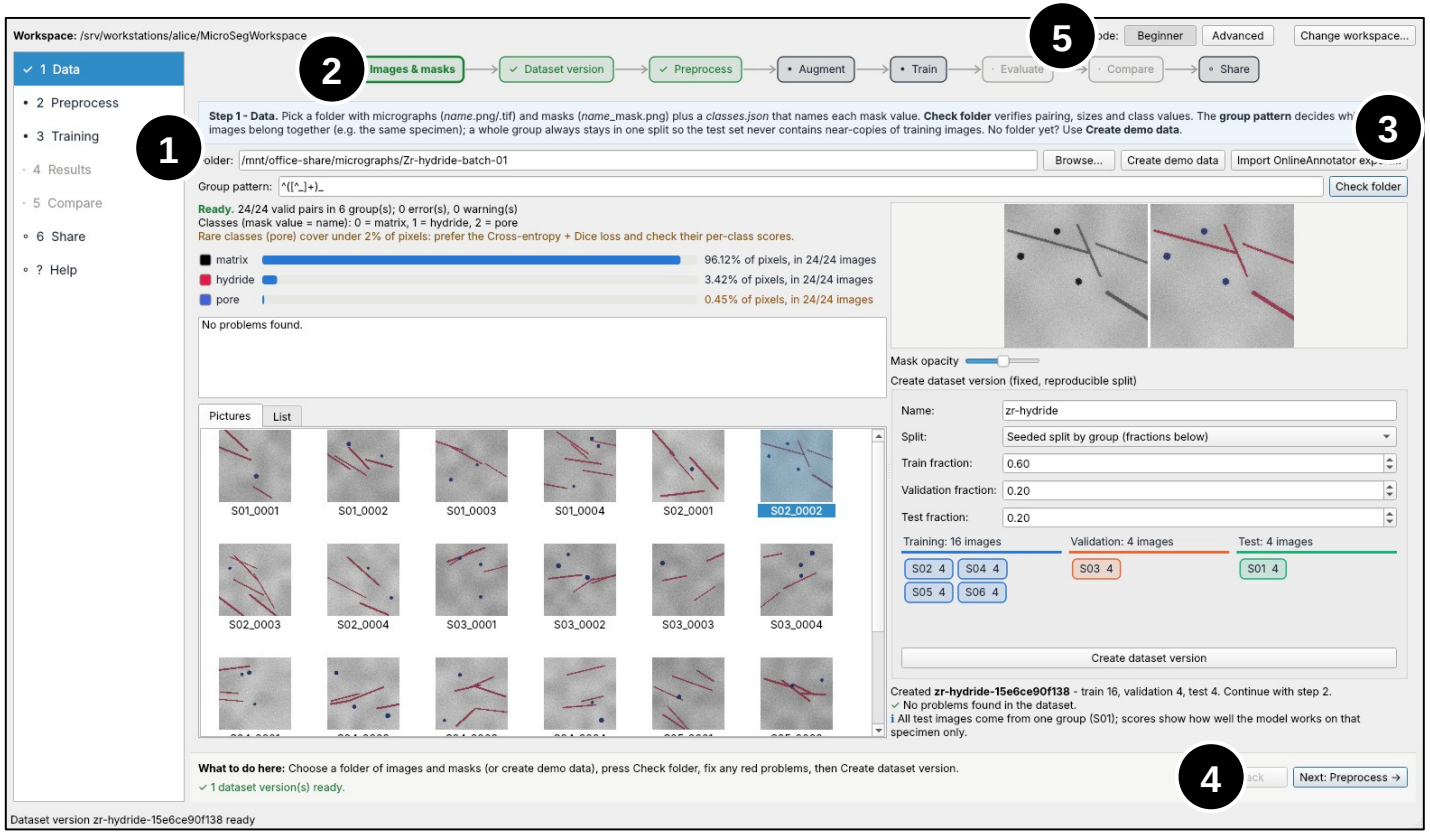
1. **Steps 1-6 on the left**
   a tick when done, ! when something needs a look

2. **Workflow map on top**
   click a stage to jump there

3. **Blue box explains the step**

4. **Footer: what to do here**
   plus Back and Next

5. **Beginner / Advanced mode**
   Beginner shows only what you need



Simulated data and network drive

Follow the map left to right; the footer always says what to do next

Step 1 · Prepare the image folder

One mask per image
S01_0001.png with S01_0001_mask.png

Mask pixel value = class number
classes.json names every value

Name starts with the specimen
S01_... keeps a specimen's images together

Or import an OnlineAnnotator export
approved annotations, provenance kept

Zr-hydride-batch-01

```
classes.json
S01_0001.png
S01_0001_mask.png
S01_0002.png
S01_0002_mask.png
...
S06_0004.png
S06_0004_mask.png
```

0	0	0	1	1	0
0	0	1	1	0	0
0	1	1	0	0	0
1	1	0	0	2	0
1	0	0	2	2	2
0	0	0	0	2	0

A mask is an image of class numbers, one per pixel

classes.json

0 = matrix
1 = hydride
2 = pore

S01_0001
^ specimen (group)

Group pattern ^([^_]+)_ takes the text before the first _ as the specimen.

All images of one specimen go to the same split.

ATTENTION
Save masks as PNG. JPEG and resizing change pixel values and silently create wrong classes.

Correct names and exact mask values are the foundation of every result

Step 1 · Check the folder

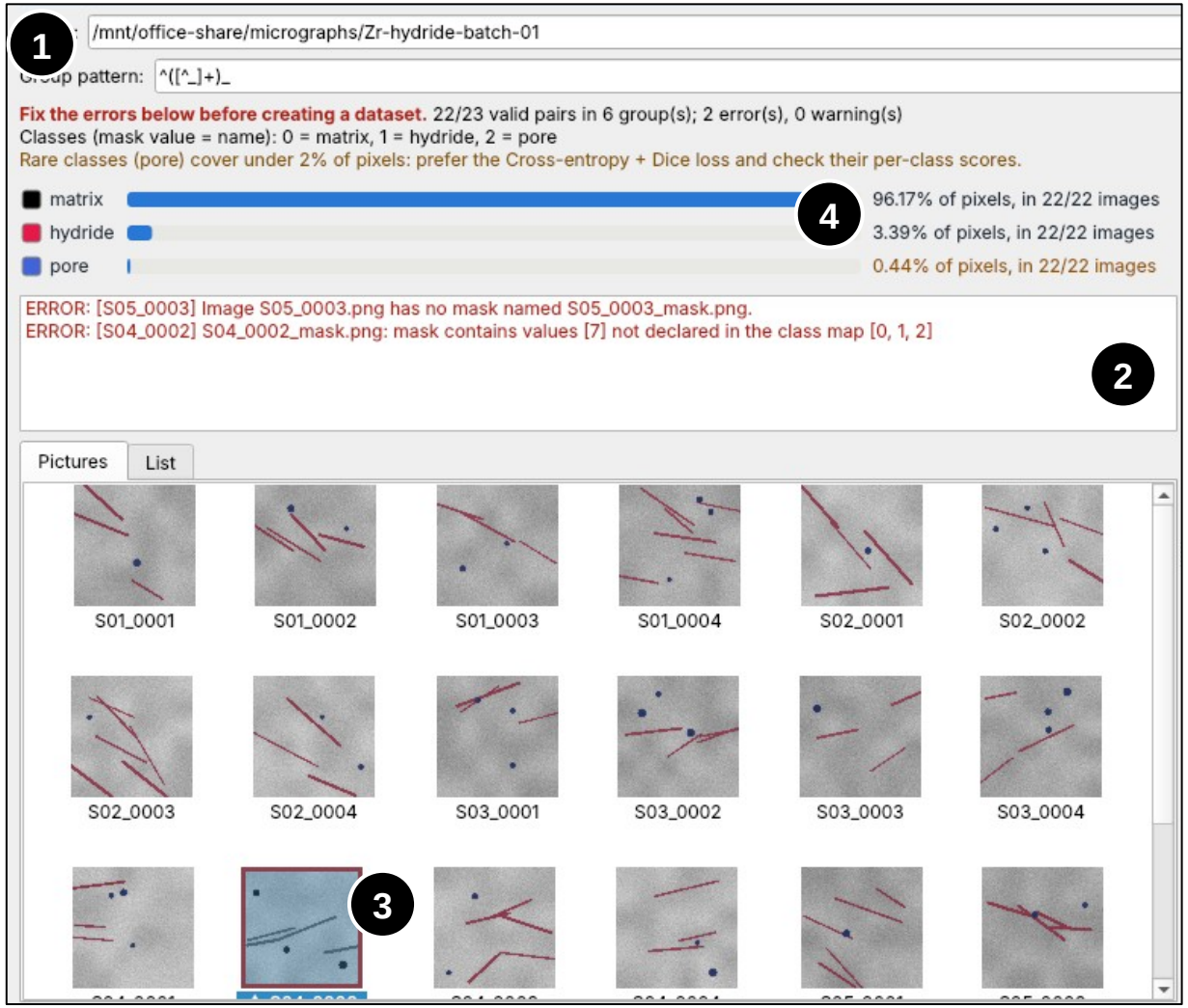
1. **Browse to the folder, press Check folder**

2. **Errors listed in plain words**
   here: a missing mask and an undeclared value 7

3. **Problem pictures framed red**
   click one to see it

4. **Class balance**
   rare classes flagged in amber

> ! **ATTENTION**
> Fix problems in the original files, then press Check folder again.



Simulated data and network drive

Fix every red problem; the app will not build a dataset from a broken folder

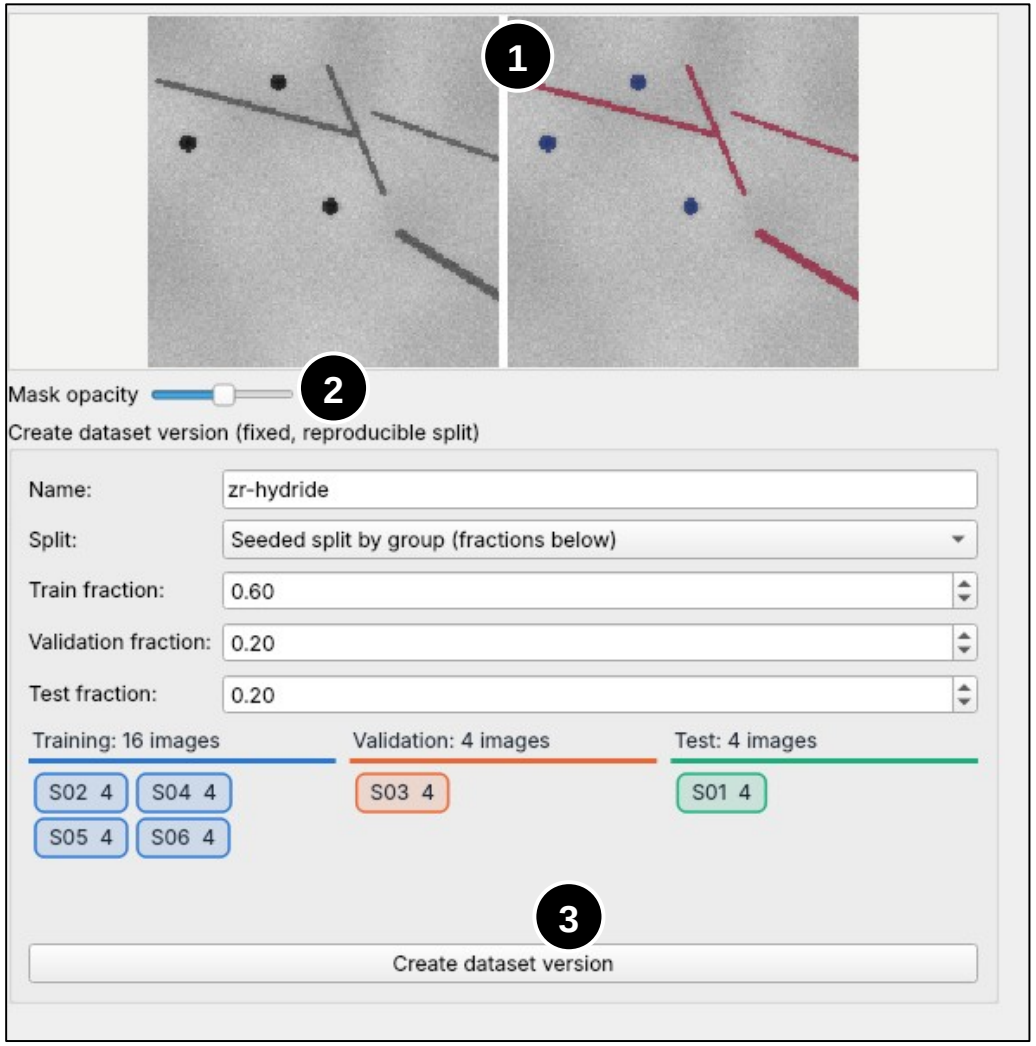
Step 1 · Split by specimen

1 Image and mask side by side
click any picture in the gallery

2 Mask opacity slider
check that masks sit exactly on the features

3 Whole specimens per split
Training learns, Validation picks the best epoch, Test is scored once

! ATTENTION
With one test specimen, test scores describe only that specimen. Use more specimens where you can.



Simulated data and network drive

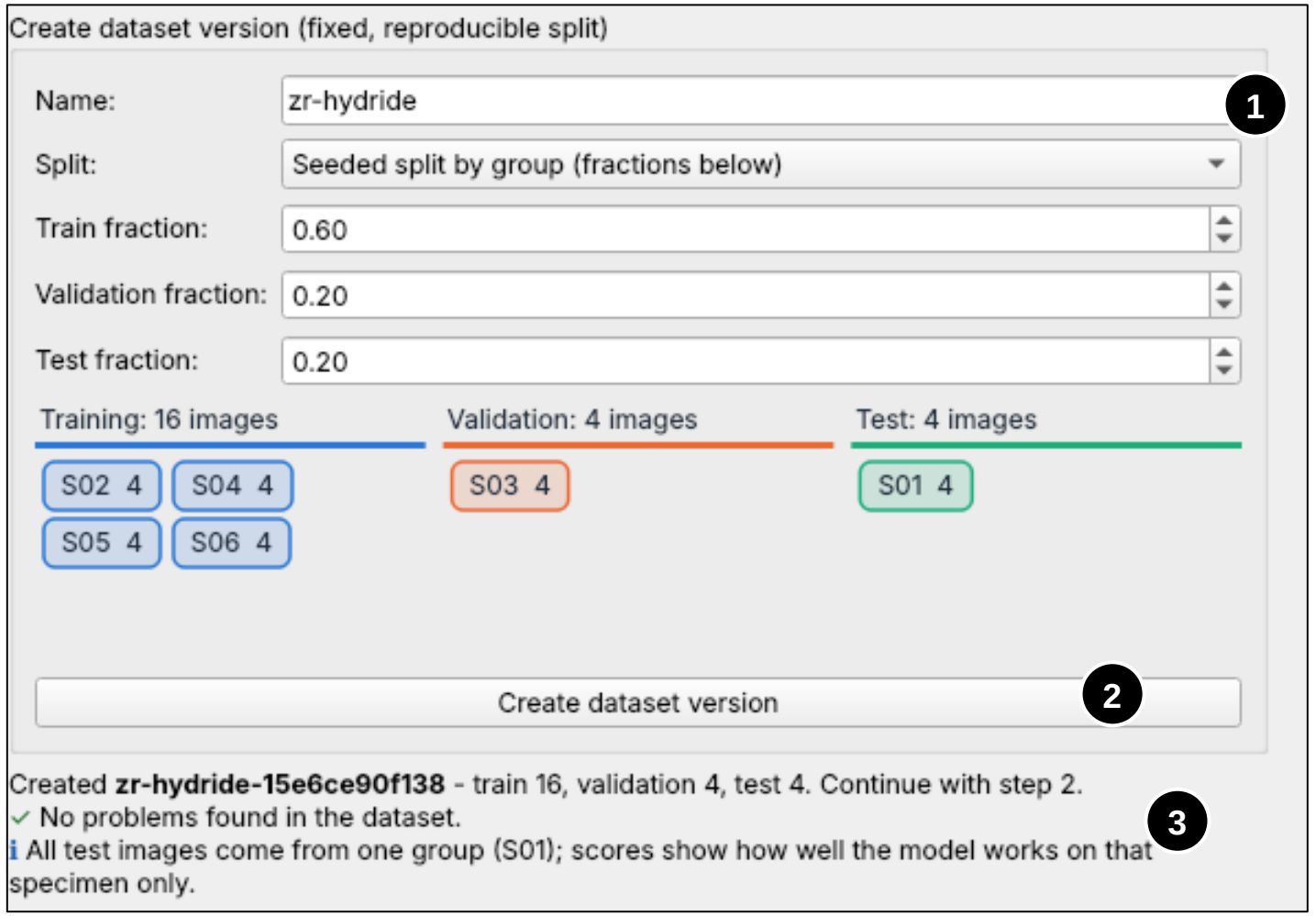
Splitting by specimen keeps the test honest

Step 1 · Create the dataset version

1. **Give the version a name**

2. **Create dataset version**
   a frozen copy with fingerprints; the same version always gives the same split

3. **Dataset check runs automatically**
   green tick, blue note, amber warning or red block



Simulated data and network drive

! ATTENTION
Adding or fixing images later? Create a new version; old results keep pointing to the old one.

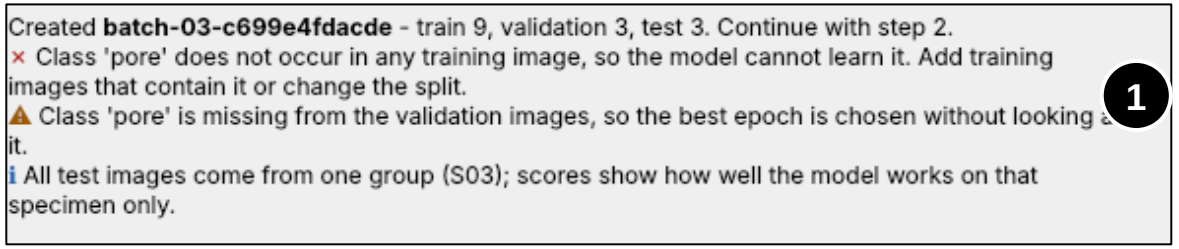
A dataset version is frozen: every result points to exactly these files

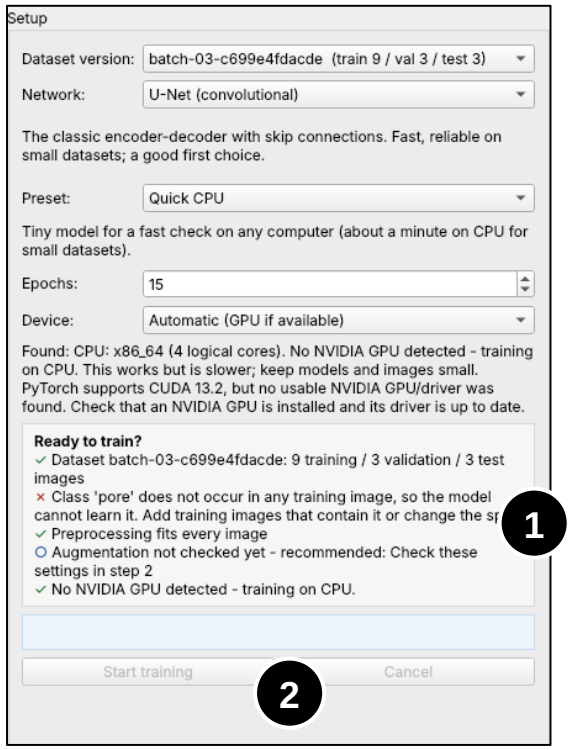
When the dataset check stops you

1 **Class missing from training**
pores occur only in specimen S03, which went to the test split

2 **Start training stays disabled**
the checklist says why

Fix the data, not the check
annotate pores in more specimens, then create a new version





Simulated example: dave's dataset

! **ATTENTION**
Amber warnings need a look; red blocks need a fix before training.

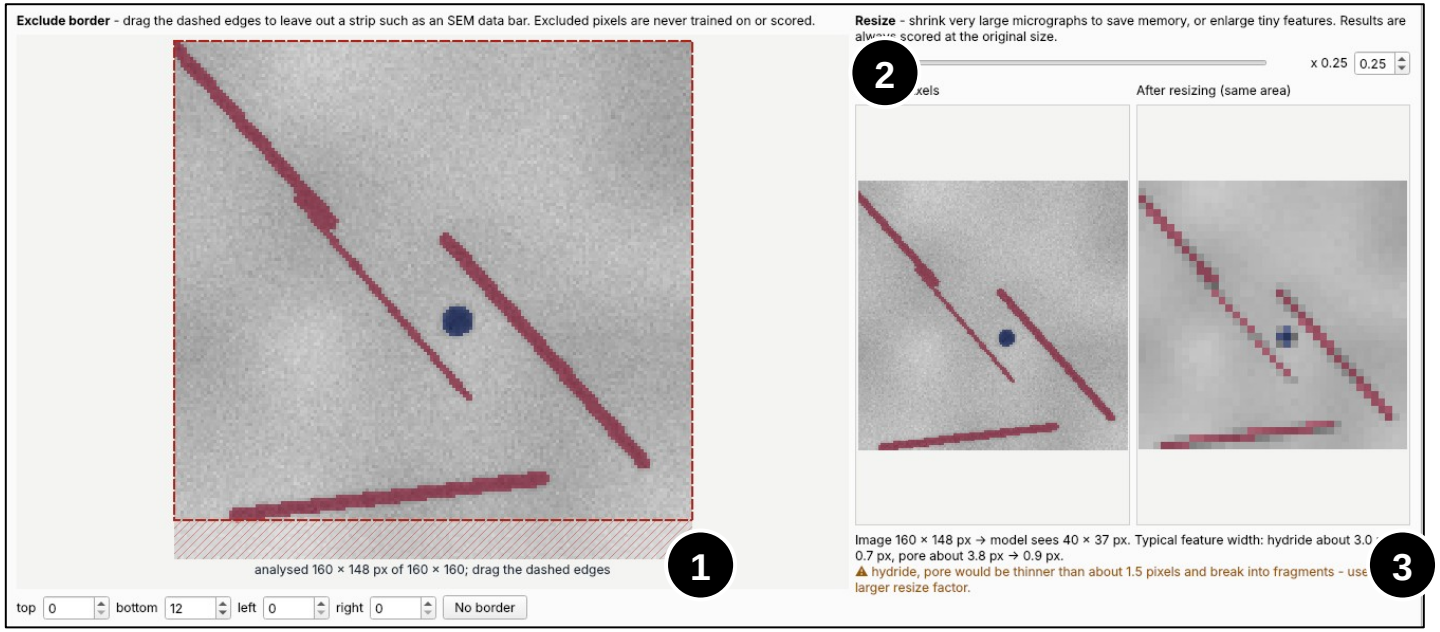
A blocked start saves hours of training a useless model

Step 2A · Preprocessing

1. **Drag the dashed edges**
   to leave out data bars or labels; excluded pixels are never trained or scored

2. **Resize only for very large images**
   before/after at true pixel size

3. **Feature-width warning**
   hydrides 3 px wide become 0.7 px at x0.25



Simulated data and network drive

> **ATTENTION**
> Keep resize at x1 unless memory forces it.
> Features thinner than ~1.5 px break apart.

Preprocess as little as possible; never shrink features away

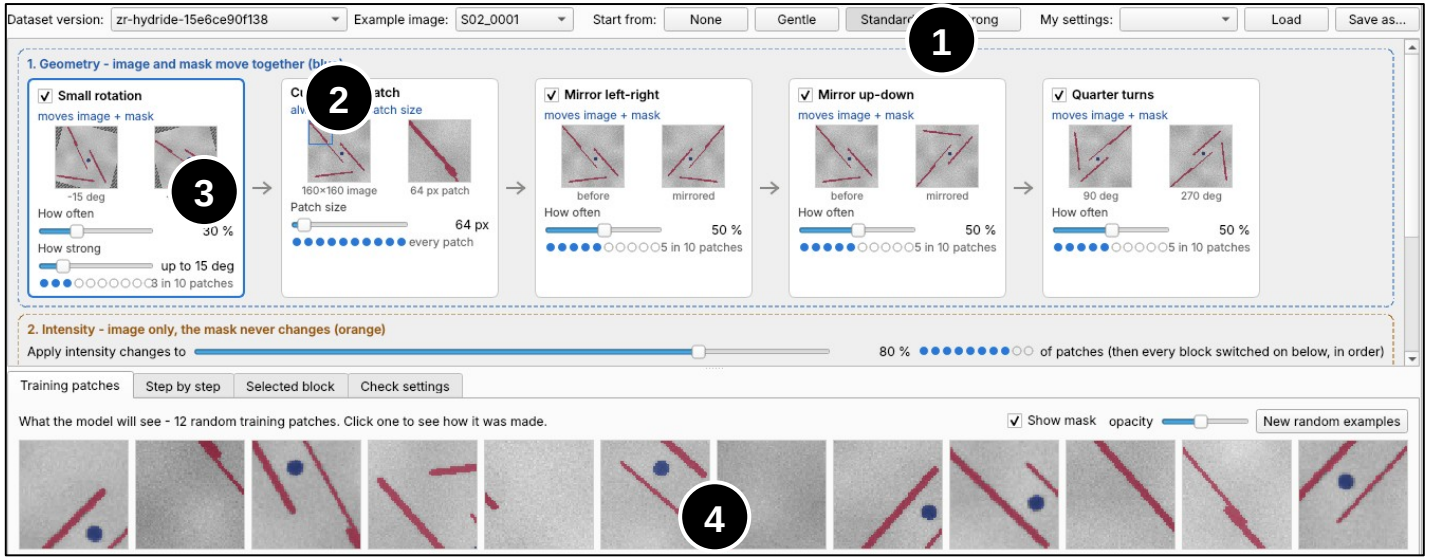
Step 2B · The augmentation pipeline

1. **Start from a preset**
   Standard suits most data

2. **Blocks run left to right**
   blue: image and mask move together; orange: image only

3. **Dots = how often a block fires**
   slider sets probability, thumbnails show weakest and strongest

4. **Real training patches**
   exactly what the model will see



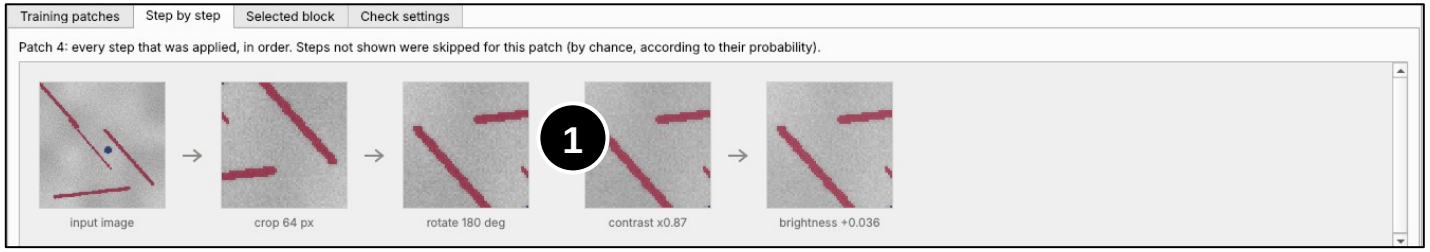
Simulated data and network drive

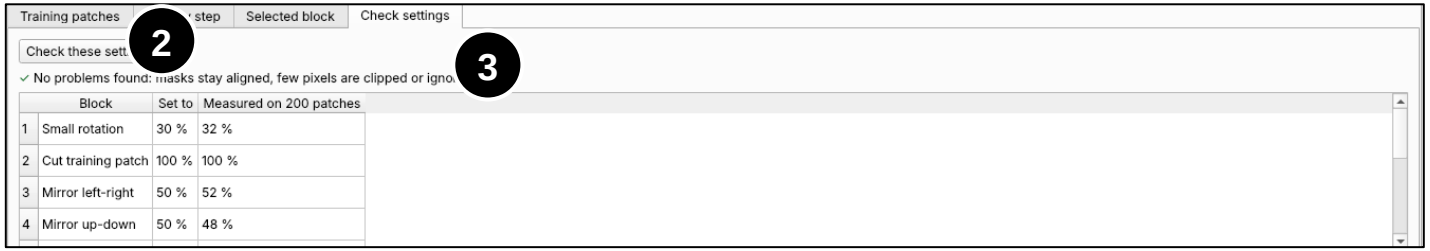
Step 2B · Check before you train

1 Step by step
one patch followed through every block

2 Check these settings
runs the real pipeline on your images

3 Findings in plain words
clipping, ignored pixels, blur against feature width, fire rates





Simulated data and network drive

! ATTENTION
Strong blur or contrast changes can erase thin features. Read every finding.

Run 'Check these settings' after every change

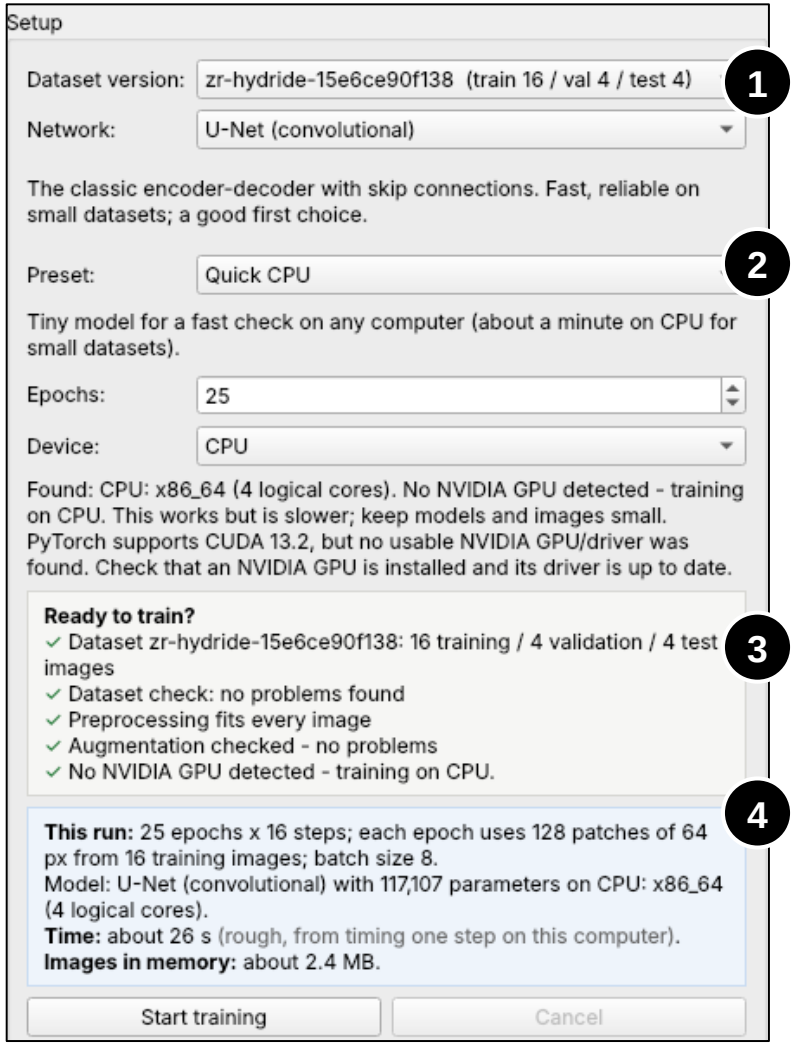
Step 3 · Set up a training run

1. **Dataset version and network**

2. **Preset**
   Quick CPU for a first try; Standard or Large GPU for real runs

3. **Ready to train?**
   all ticks, or amber items you have read

4. **This run**
   epochs, model size, time and memory, measured on this PC



Simulated data and network drive

> **!** **ATTENTION**
> No NVIDIA GPU found means training runs on the CPU: slower, but it works.

Read the checklist and the run card before pressing Start

Choosing a network

U-Net: first choice
convolutional, fast, reliable on small datasets

SegFormer-lite: transformer
sees wider context; needs more images and epochs

Train both on the same dataset version
then compare them in step 5

More networks can be added later
as plugins, without changing the app

	U-Net	SegFormer-lite
Type	Convolutional encoder-decoder	Vision transformer, MLP decoder
Strength	Fine detail, small data	Long-range context
Needs	Few images, short training	More images, more epochs
Patch size	multiple of 2^depth	multiple of 32
Parameters (quick preset)	117,107	462,211
Demo test feature IoU	0.995	0.860
Epochs in the demo	25	40

Demo: synthetic data, 16 training images, CPU; U-Net 25 epochs, SegFormer-lite 40 epochs. Your numbers will differ.

Start with U-Net; try the transformer when you have more data

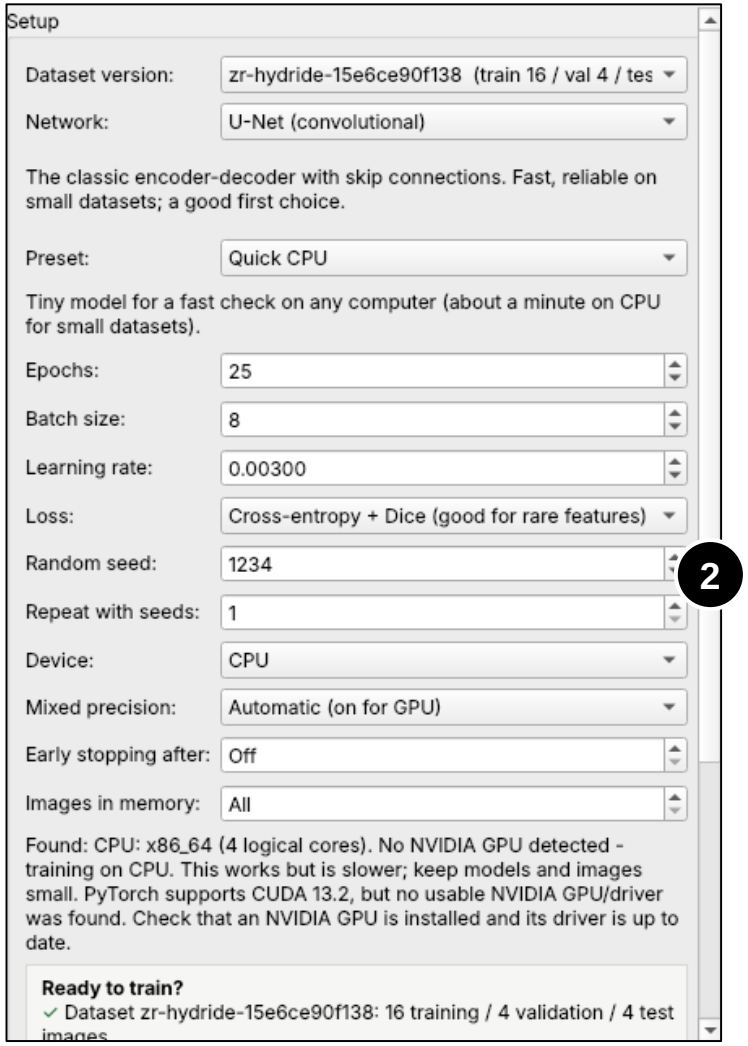
Beginner and Advanced mode

Switch at the top right
remembered for your workspace

2 **Advanced adds expert settings**
batch, learning rate, loss, seed, repeats, precision, early stopping

Beginner keeps the essentials
dataset, network, preset, epochs, device

! **ATTENTION**
Change one expert setting at a time, and compare the runs in step 5.

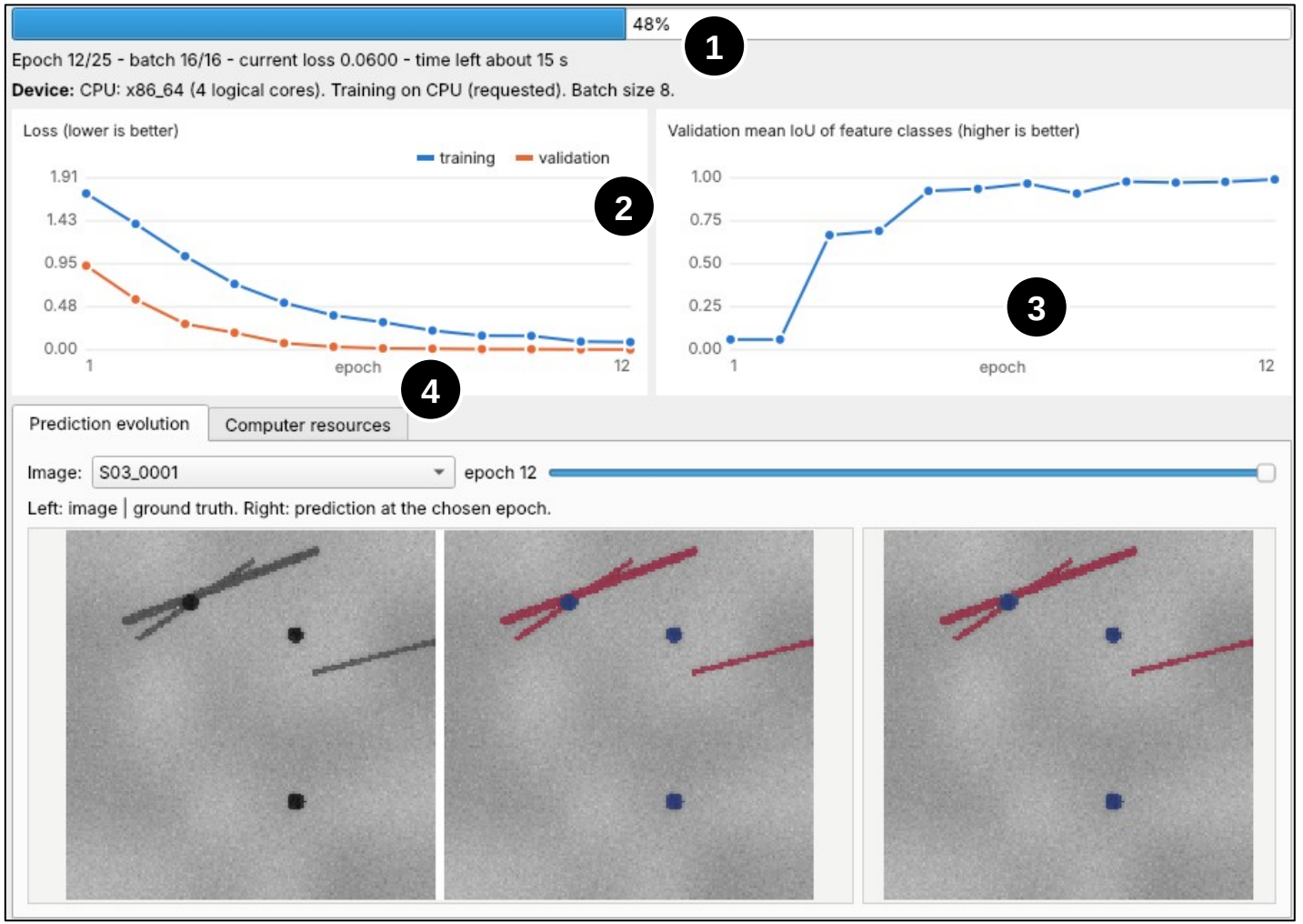


Simulated data and network drive

Stay in Beginner mode until a result tells you what to change

Step 3 · Watching training

1. **Progress and time left**
2. **Loss falls**
   training (blue) and validation (orange)
3. **Validation IoU rises, then levels off**
   the best epoch is kept automatically
4. **Prediction evolution**
   the same image, epoch by epoch



Simulated data and network drive

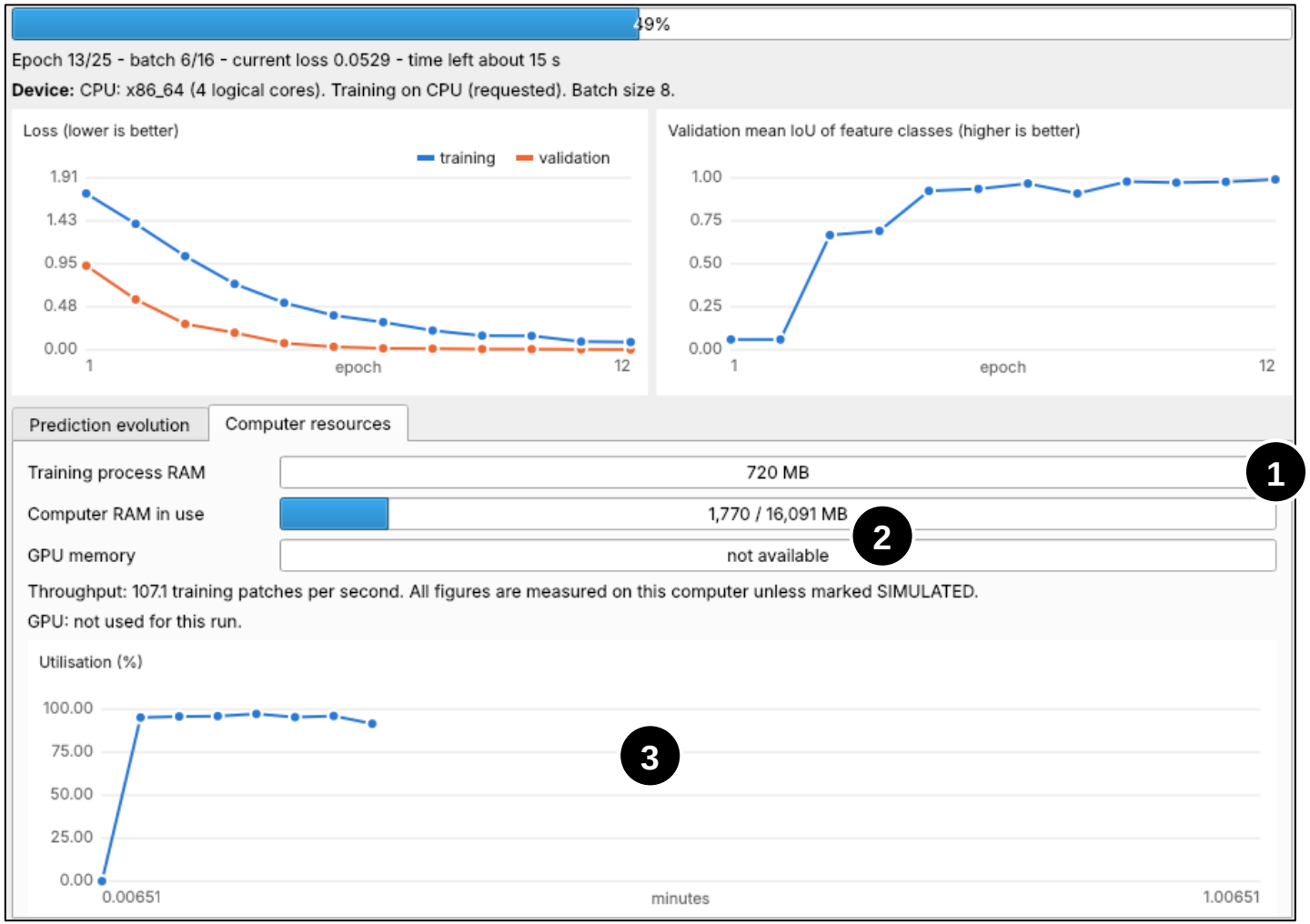
ATTENTION
Training loss falling while validation IoU drops = overfitting. Cancel is safe; resume later in step 4.

Judge training by validation curves and images, not by loss alone

Step 3 · Computer resources

1. **Memory of the run and the PC**
   and GPU memory when an NVIDIA GPU is used
2. **Measured speed**
3. **Utilisation over time**

Out of memory?
the run stops cleanly and suggests a smaller batch or patch



CPU-only sandbox: GPU memory not available

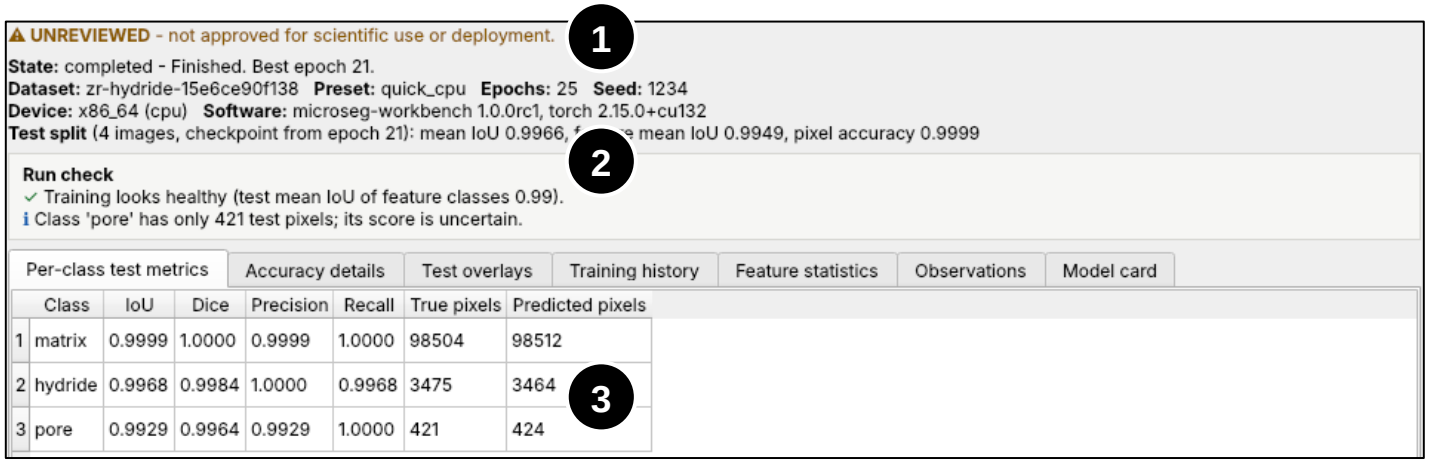
Know whether you are on the GPU or the CPU before a long run

Step 4 · Results and run check

1. **Status banner: UNREVIEWED**

2. **Run check in plain words**
   overfitting, under-training, missed classes, thin evidence

3. **Per-class test scores**
   on the test split, never used for training



Simulated data and network drive

! **ATTENTION**
Judge by the feature classes (hydride, pore). Matrix dominates the pixels, so overall accuracy flatters.

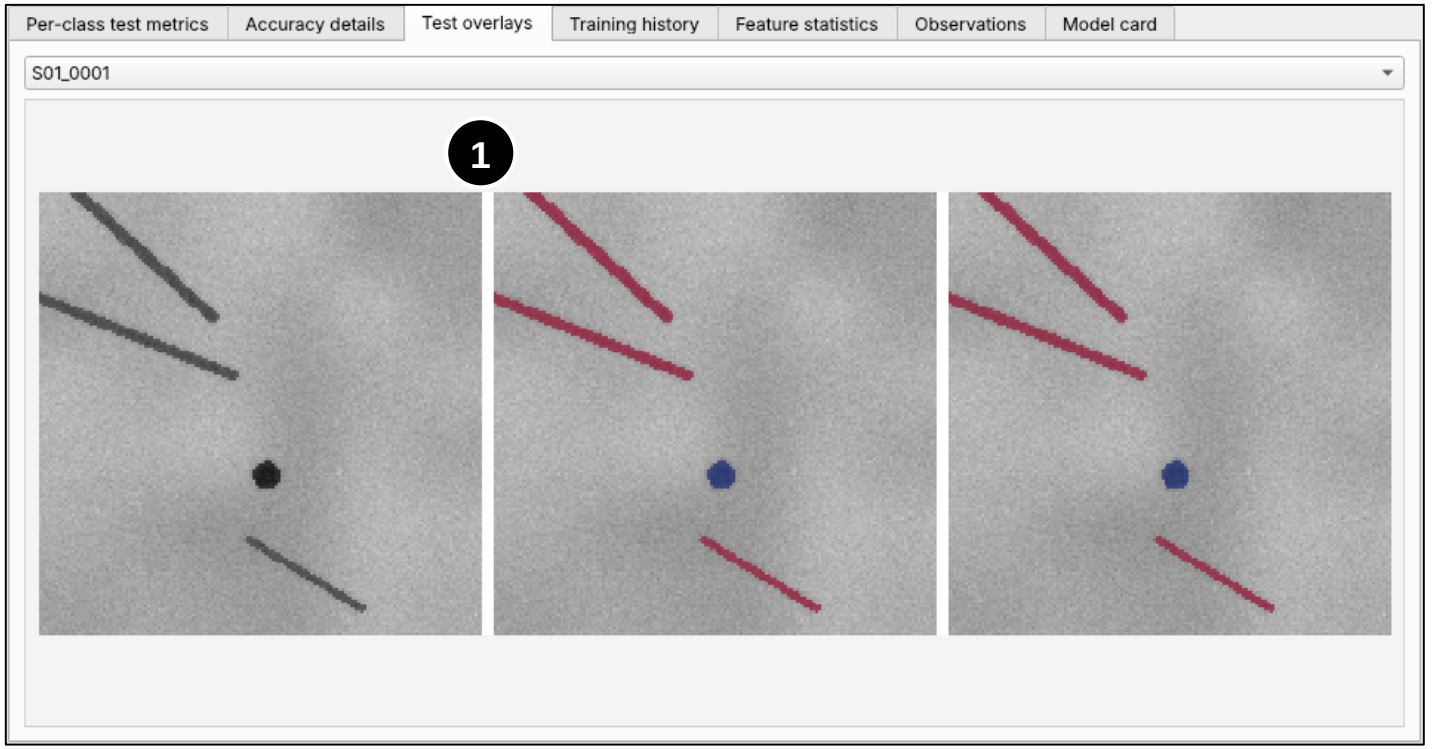
Results stay unreviewed until a colleague approves them

Step 4 · Look at the overlays

1. **Every test image**
   ground truth next to prediction

Look for
broken thin hydrides, missed or merged pores, false edges at borders

Export report
one self-contained HTML file to attach or archive



Simulated data and network drive

A good score can hide systematic errors; always look

Step 4 · Accuracy details

1 Boundary F1
share of edges within 1 px of the true edge; 1 = every edge in place

2 True vs predicted area %
a clear difference means biased phase fractions

3 Weakest test images
look at their overlays first

4 Where pixels go
which classes are confused with which

! ATTENTION
Before reporting area fractions, compare them with manual measurements on images not used for training.

Per-class test metrics | Accuracy details | Test overlays | Training history | Feature statistics | Observations | Model card

Boundaries and area fractions (1)

Class	Boundar	oundary recision	Boundary recall	True area %	Predicted area %	Difference (points)	Mean per-image area error (points)
matrix	0.9934	0.9995	0.9873	96.20	96.03	-0.17	0.17
hydride	0.9899	1.0000	0.9799	3.39	3.56	+0.17	0.19
pore	1.0000	1.0000	1.0000	0.41	0.41	-0.00	0.01

(2)

Boundary F1: share of boundary pixels within 1 px of the true boundary (1 = all edges in place). Area %: share of evaluated pixels; the difference shows whether the model systematically over- or under-segments a phase.

Weakest test images (of 4) (3)

Image	Feature mean IoU	Pixel accuracy	hydride area % true / predicted	pore area % true / predicted
S01_0003	0.8242	0.9940	2.24 / 2.41	0.37 / 0.35
S01_0004	0.8391	0.9880	4.72 / 4.68	0.53 / 0.55
S01_0002	0.8686	0.9921	3.74 / 4.00	0.43 / 0.43
S01_0001	0.9233	0.9958	2.87 / 3.15	0.31 / 0.31

Look at the overlays of these images first. All per-image scores: `evaluation/per_image.csv`.

Where pixels go (% of each true class) (4)

	predicted matrix	predicted hydride	predicted pore
true matrix	99.53	0.46	0.02
true hydride	7.94	91.91	0.14
true pore	4.99	0.00	95.01

Simulated data and network drive

Check boundaries and area fractions before you measure with a model

Step 4 · Observations and model card

1 Observations

what you checked, what the model is and is not good for

Notes cannot be edited

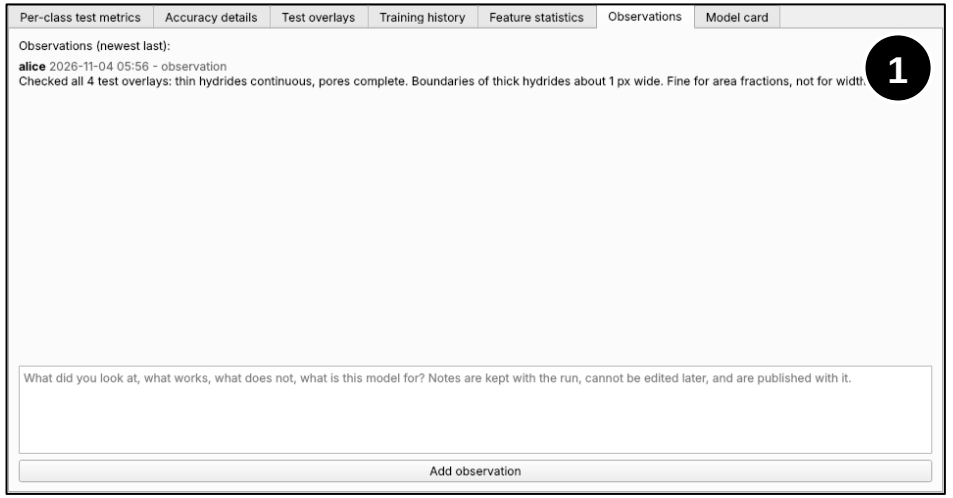
add a new note to correct one

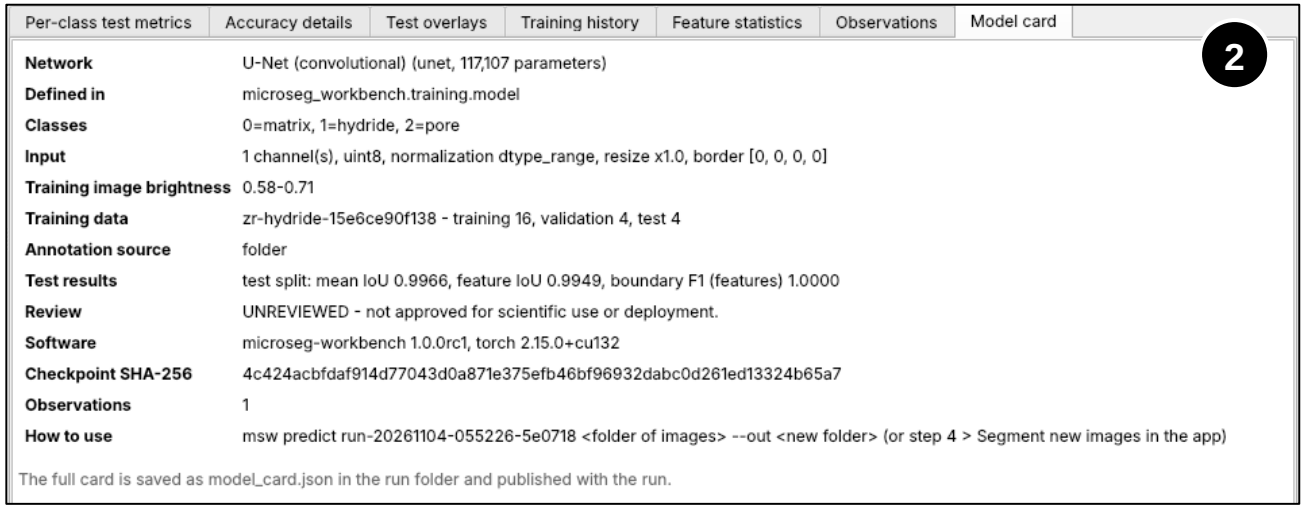
2 Model card

network, classes, input, training data, scores, software, checkpoint fingerprint

! ATTENTION

Publishing in step 6 needs at least one observation.



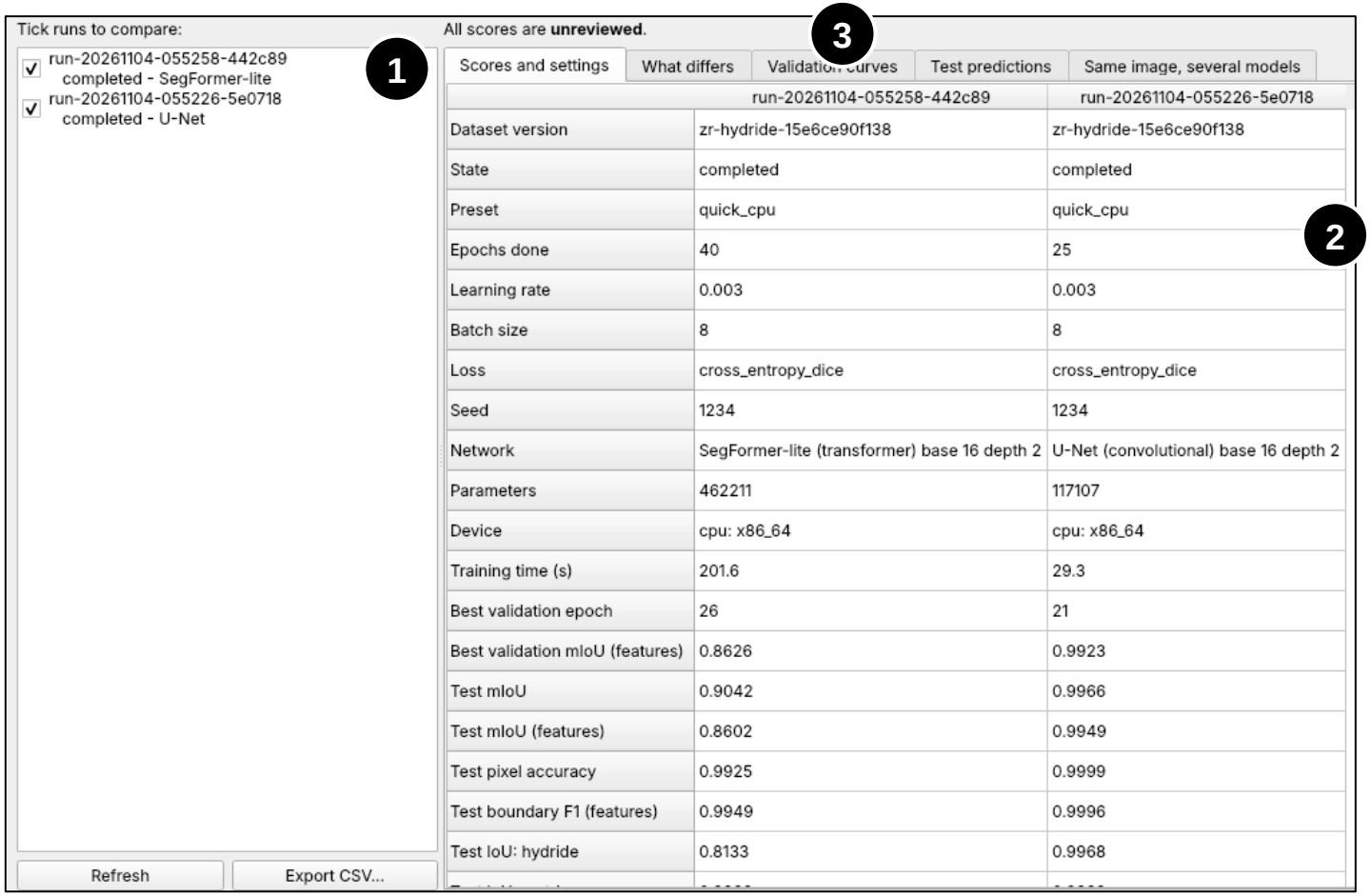


Simulated data and network drive

Write down what you checked; colleagues rely on it

Step 5 · Compare runs

1. **Tick the runs to compare**
2. **Scores and settings side by side**
   'What differs' lists only the changed settings
3. **Validation curves, test predictions**



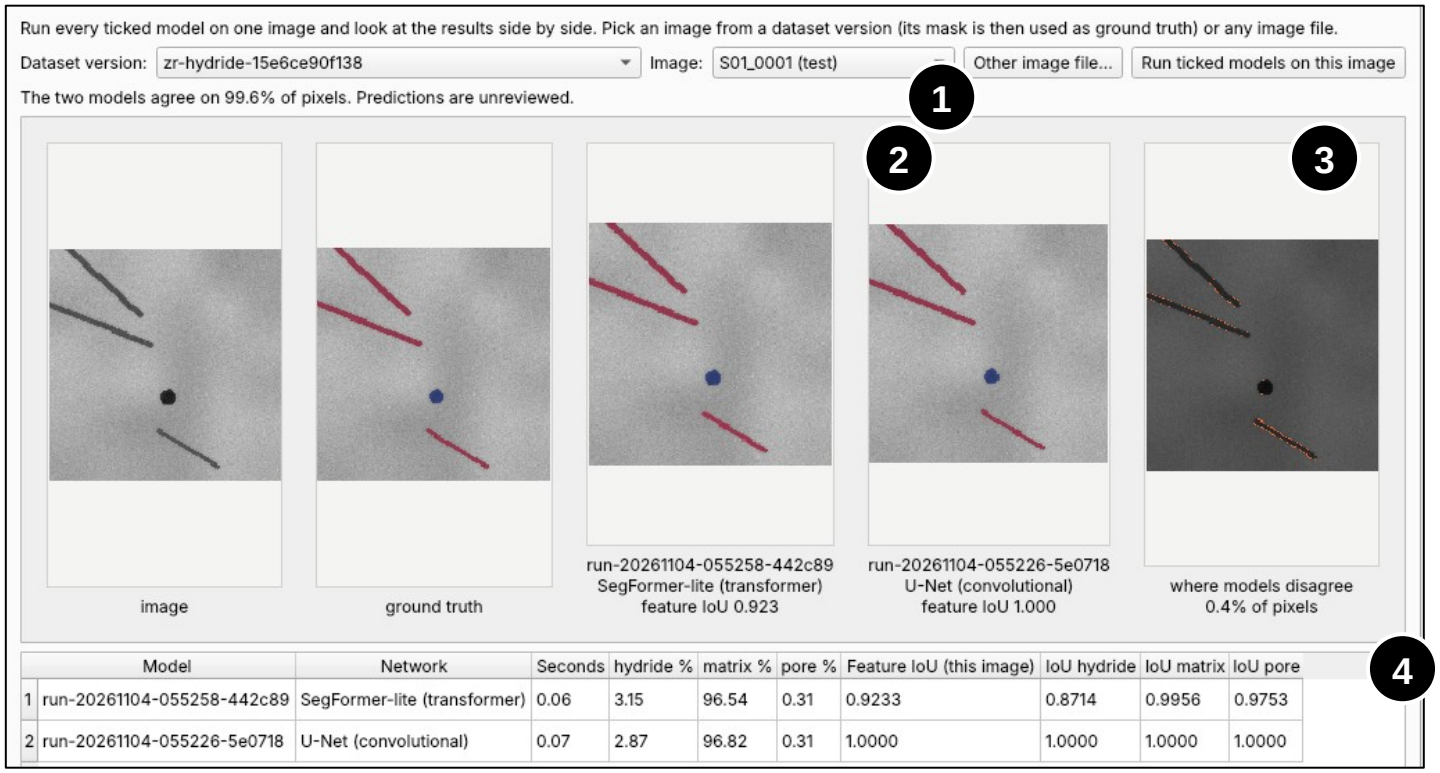
Simulated data and network drive

> **!** **ATTENTION**
> Rank runs only on the same dataset version. Repeat with seeds (Advanced) to see how much is luck.

Compare like with like: same dataset version, several seeds

Step 5 · Same image, several models

1. **Choose a dataset image or any file**
   dataset images bring their ground truth
2. **Each model's segmentation**
   U-Net and SegFormer-lite side by side
3. **Where models disagree**
   orange pixels
4. **Scores, class fractions, time**



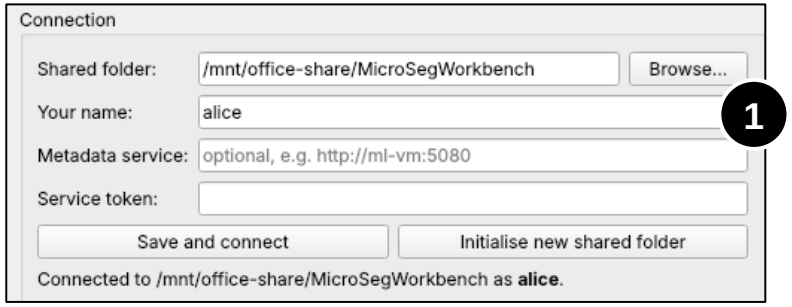
	Model	Network	Seconds	hydride %	matrix %	pore %	Feature IoU (this image)	IoU hydride	IoU matrix	IoU pore
1	run-20261104-055258-442c89	SegFormer-lite (transformer)	0.06	3.15	96.54	0.31	0.9233	0.8714	0.9956	0.9753
2	run-20261104-055226-5e0718	U-Net (convolutional)	0.07	2.87	96.82	0.31	1.0000	1.0000	1.0000	1.0000

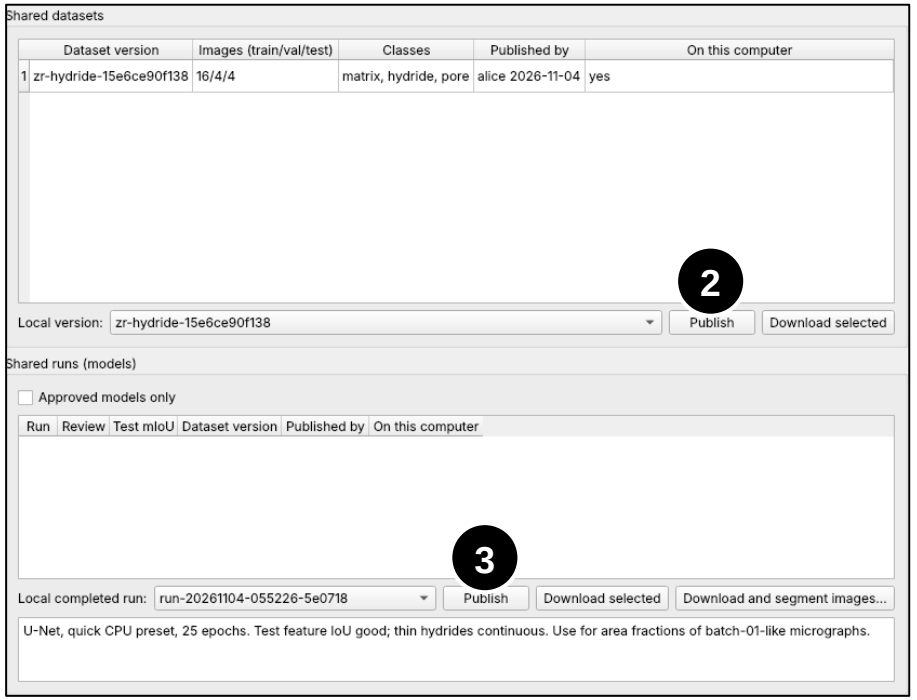
Simulated data and network drive

Disagreement shows where to look more closely

Step 6 · Connect and publish

1. **Shared folder path and your name**
   once; Save and connect
2. **Publish the dataset version first**
3. **Write observations, publish the run**
   files verified, deduplicated, immutable





Simulated data and network drive

> **!** **ATTENTION**
> Published items are never overwritten. To improve a model, train and publish a new run.

Publishing makes your model available, with your notes attached

Step 6 · Review by a colleague

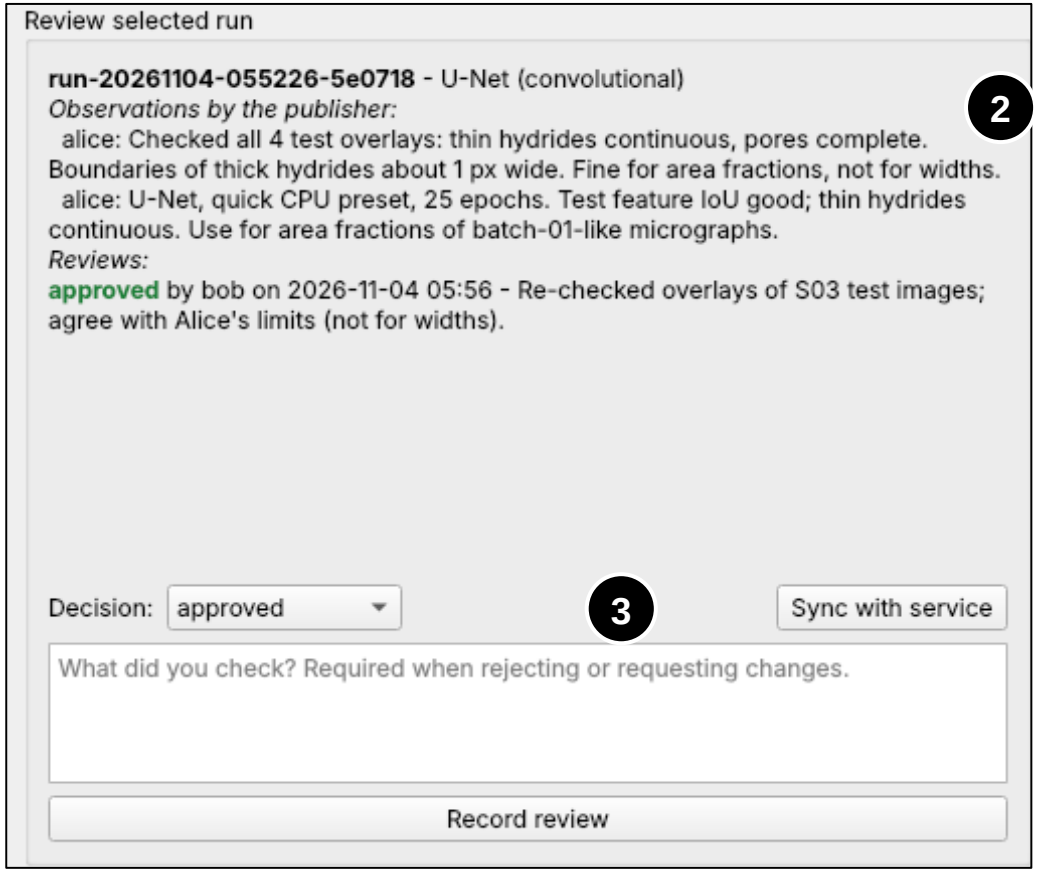
Download the run
with its dataset, verified

2. **Read the observations, check the overlays**

3. **Decide and comment**
approved, needs changes or rejected

4. **Status for everyone**
UNREVIEWED becomes APPROVED





Simulated data and network drive (bob's PC)

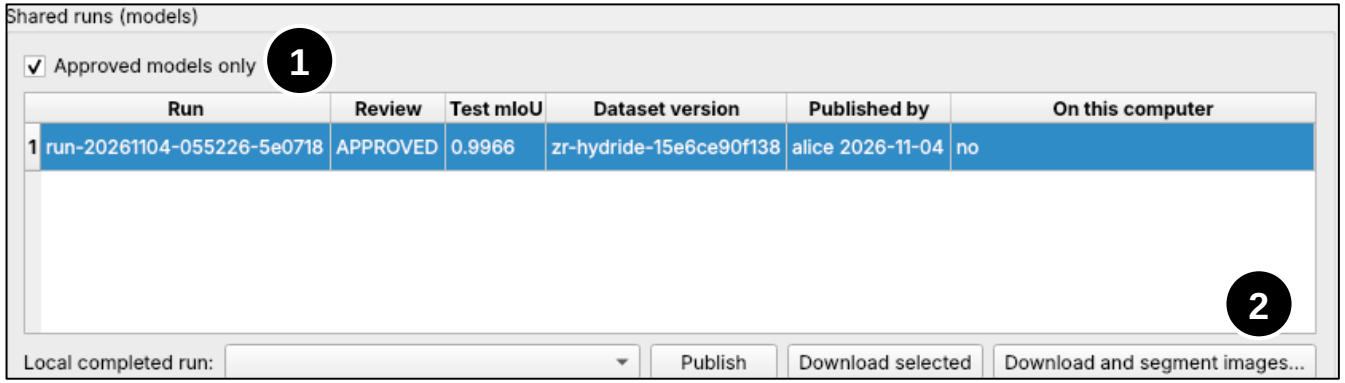
Approval must come from someone other than the publisher

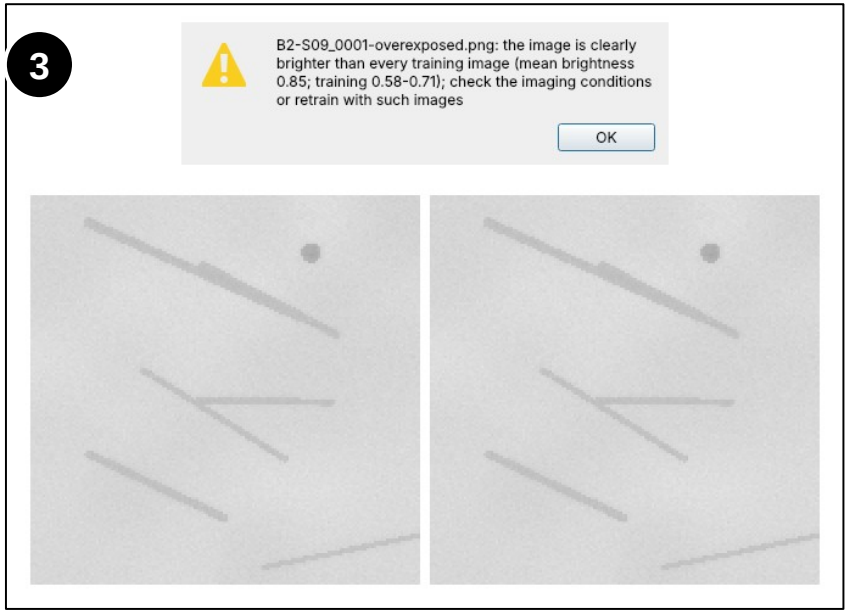
Fetch and use an approved model

1. **Approved models only**

2. **Download and segment images**
   pick a folder; model and data are fetched, then applied

3. **Warning for unlike images**
   here: an over-exposed micrograph, on which the model found no hydrides at all





Warning, then the output overlay: image | prediction (nothing found)

! ATTENTION
Predictions are unreviewed. Check images with warnings, or retrain with such images.

Heed input warnings: a model knows only images like its training images

The same work from the command line

For repeat or batch work
same checks, same files, same provenance

msw models
every model here and on the shared folder

msw predict
downloads a shared model first

msw selftest
checks this computer and the shared folder

msw --help
lists every command

```
> msw models
run-20261103-093320-629a63  [APPROVED]  U-Net (convolutional)
   data zr-hydride-15e6ce90f138  test feature IoU 0.9949
   (here + shared)

> msw predict run-20261103-093320-629a63
      S:\micrographs\Zr-hydride-batch-02-unlabelled
      --out C:\Users\carol\results\batch-02
Downloading model ... (only if not here yet)
[1/9] B2-S01_0001.png
  ...
[9/9] B2-S09_0001-overexposed.png
WARNING B2-S09_0001-overexposed.png: the image is
  clearly brighter than every training image (mean
  brightness 0.85; training 0.58-0.71); check the
  imaging conditions or retrain with such images
9 prediction(s) written to C:\Users\carol\results\
  batch-02  [UNREVIEWED]

> msw compare-image IMAGE RUN RUN --out side.png
> msw check-run RUN      > msw report RUN      > msw selftest
```

Everything in the app is also available as msw commands

Ten things that need your attention

Before you trust a result

Before you share it

Before you use someone else's model

Print this page
and keep it next to the workstation

1. Masks are PNG with exact class values; classes.json names every value
2. File names start with the specimen, so splits keep specimens together
3. Fix every red problem; read every amber warning
4. Never edit files inside the workspace or shared folder by hand
5. Do not resize features below ~1.5 px
6. Run 'Check these settings' after changing augmentation
7. Judge by feature IoU and the overlays, not overall accuracy
8. Compare runs only on the same dataset version (and several seeds)
9. Write observations before publishing; never approve your own run
10. Unreviewed is not approved; check images that raise input warnings

Most mistakes are caught by the app; the rest by your eyes

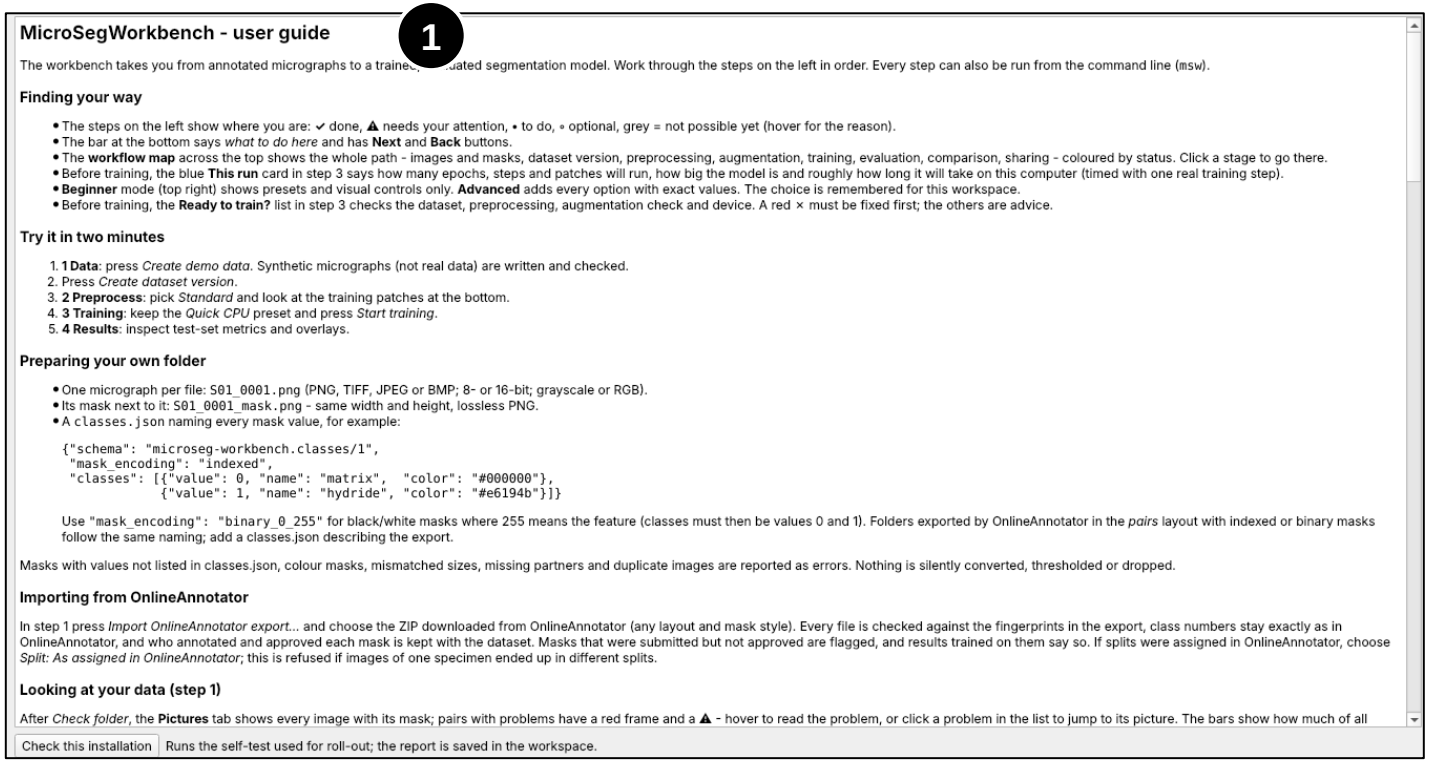
Getting help

1. **? Help page**
   every step explained, with the matching commands

Hover for tooltips
on every button and setting

Error messages say what to do
not just what went wrong

Administrators
README, docs/DEPLOYMENT.md, docs/SPECIFICATION.md



Simulated data and network drive

When unsure: read the step's help, then ask a colleague